

Review

The development and domestication of maize ear

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Abstract Maize is amongst the most agriculturally and economically important crops to human beings. It was domesticated from a wild relative called teosinte. During domestication, maize has experienced drastic morphological transformations, such that it produces fewer ears, each of which bears many more kernels covered by soft and reduced glumes. The striking differences between maize and teosinte make the origin of maize ear a fascinating question, which has been fiercely and actively debated for more than a century. Over the past few decades, the discovery of numerous key genes and genetic pathways has greatly deepened our understanding of the mechanisms underlying maize ear development and domestication. In this review, by providing an overview of the morphogenetic processes of maize and teosinte ears, and the molecular mechanisms of maize ear development, we highlight key morphodynamical distinctions between maize and teosinte ears. By recapitulating historical accounts and summarizing recent advances regarding maize domestication, we present the current understanding and propose a model for the origin of maize ear.

Key words: domestication, ear, inflorescence, maize, teosinte.

1 Introduction

Maize (*Zea mays* L. ssp. *mays*) has been exploited by humankind as a major ingredient of food, a component of livestock feed, and more recently, a source of renewable energy. Its worldwide popularity and importance are a result of the alteration of many key traits during domestication, including the loss of photoperiod sensitivity (Yang et al., 2013; Huang et al., 2018), the enhancement of cold tolerance (Zeng et al., 2025), the change of nutrients (Whitt et al., 2002), and more importantly, the increase of yield (Chen et al., 2022). The significant increase in yield is directly related to the modification of the ear (female inflorescence) architecture. During domestication, maize ear has experienced tremendous morphological transformation. A suite of phenotypes, such as ear size, rank number, kernel number, and glume size and texture, has been significantly altered from its wild relative, teosinte. A maize ear is seated at the leaf axil and is enveloped by several sterile protecting leaves called husks. A mature ear is nearly cylindrical, with at least eight rows (derived from four ranks, each of which generates two rows) of kernels arranged in an approximately whorled pattern on the central fibrous rachis (Fig. 1A). A mature kernel is subtended by a pair of minute and papery glumes, which renders the kernel exposed and hence eases the harvesting process (Fig. 1A).

The unique appearance of maize ear is profoundly different from those of inflorescences of any other Poaceae species, so much so that it was once considered a monstrosity (Beadle, 1972). This has stimulated considerable controversies and debates historically regarding its origin (Harshberger, 1896; Mangelsdorf & Reeves, 1938, 1939; Beadle, 1939, 1972, 1980, 1981; Iltis, 2000; Doebley, 2001). Despite recent breakthroughs in revealing the ancestry of maize (Yang et al., 2023), an in-depth understanding of the molecular and genetic basis of the domestication of maize ear is required. In this review, we first provide an overview of the morphogenetic process and molecular mechanisms of maize ear development, then revisit historical debates regarding the domestication of maize and highlight recent progress in identifying and analyzing key domestication genes, and lastly synthesize current insights to propose a model for the origin of maize ear.

2 Molecular mechanisms underlying maize ear development

The morphogenesis of maize ear follows a highly stereotypical series of developmental events (Cheng et al., 1983; McSteen et al., 2000; Tanaka et al., 2013) (Fig. 1B). Upon the conversion from an axillary meristem (AM), the female

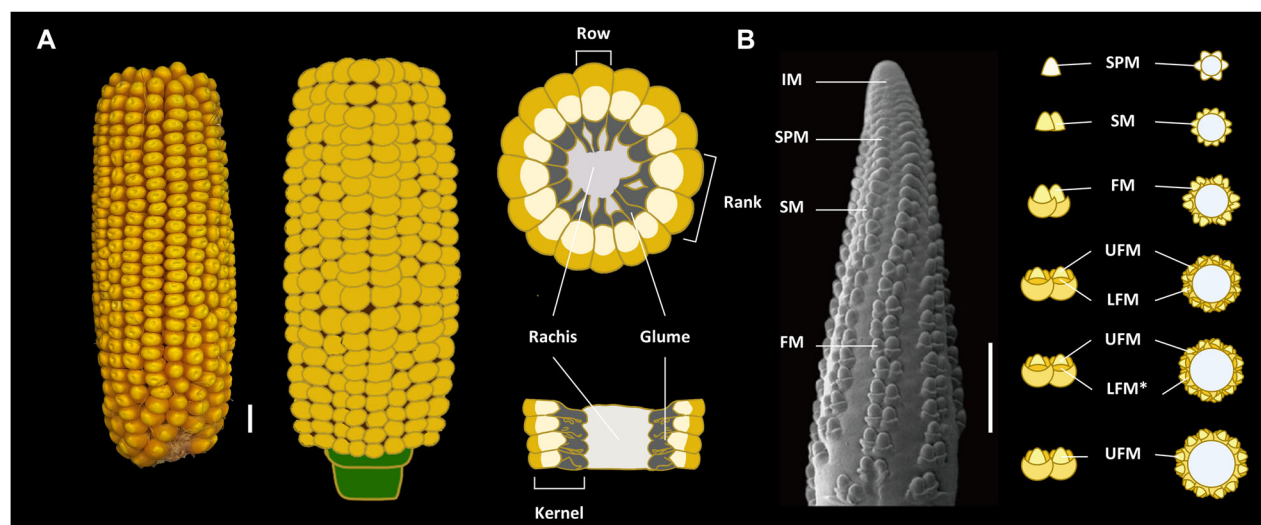


Fig. 1. Development of maize ear. **A**, Real and cartoon images of a mature maize ear, with cross and partial longitudinal sections. The false grey color of the glume is for better visualization. Bar = 1 cm. **B**, Scanning electron micrograph of a developing maize ear, with schematic meristem identities. Bar = 500 μ m. The asterisk indicates an aborting organ. The scanning electron micrograph is modified from Shen et al. (2023) with permission. Abbreviations: FM, floral meristem; IM, inflorescence meristem; LFM, lower floral meristem; SM, spikelet meristem; SPM, spikelet pair meristem; UFM, upper floral meristem.

inflorescence meristem (IM) produces numerous spikelet pair meristems (SPMs) acropetally in a spiral pattern (Chuck et al., 2010). The SPM is determinate, giving rise to two spikelet meristems (SMs), the pedicellate first and the sessile slightly later (Vollbrecht & Schmidt, 2009). After the formation of a pair of glumes, each SM forms a lower and then an upper floral meristem (FM) (Chuck et al., 1998; Yang et al., 2022). The upper FM produces a floret, which consists of two bract-like structures (lemma and palea) and floral organs (lodicules, arrested stamens, and carpels), whereas the lower FM aborts. The developmental process of maize ear is orchestrated by numerous highly coordinated genes regulating meristem identity maintenance, transition, and determinacy (Table 1).

The most extensively investigated genetic program in maize ear development is the WUSCHEL (WUS)-CLAVATA (CLV) signaling feedback loop (Somssich et al., 2016; Fletcher, 2018; Wu et al., 2018) (Fig. 2A). Maize has two WUS paralogs with different expression patterns in the female IM. *ZmWUS1* is expressed in the organizing center, whereas *ZmWUS2* is expressed more broadly and evenly within the female IM, although often with a slight preference for the epidermis (Nardmann & Werr, 2006; Je et al., 2016). *ZmWUS1* and *ZmWUS2* co-bind many genes regulating meristem function, indicating that they may function in a redundant manner (Dong et al., 2023). Misregulation of WUS genes causes ear abnormalities. For instance, the dominant *barren inflorescence3* (*bif3*) mutant harbors a tandem duplicate copy of *ZmWUS1*, which is ectopically expressed within the female IM roughly corresponding to a hollow sphere, resulting in the formation of ball-shaped ears and a limited number of SPMs (Chen et al., 2021, 2024). *ZmWUS1* activity is repressed by a signaling peptide *ZmCLV3/EMBRYO SURROUNDING REGION7* (*ZmCLE7*), encoded by a putative CLV3 ortholog (Je et al., 2016). *ZmCLE7* is perceived by

various receptors, including THICK TASSEL DWARF1 (TD1), a receptor-like kinase encoded by the CLV1 ortholog (Bommert et al., 2005; Lunde & Hake, 2009), and FASCIATED EAR2 (FEA2), a receptor-like protein (Taguchi-Shiobara et al., 2001; Bommert et al., 2013a) that interacts with COMPACT PLANT2 (CT2), the α -subunit ($G\alpha$) of a heterotrimeric GTP-binding protein (Bommert et al., 2013b). Defects of *ZmCLE7* signaling result in a class of similar phenotypes: enlarged, fasciated, ridged, bifurcated, or otherwise misshapen ears often carrying extra and/or disorganized kernel rows (Rodriguez-Leal et al., 2019; Liu et al., 2021). Another signaling peptide, *ZmFON2-LIKE CLE PROTEIN1* (*ZmFCP1*), also suppresses *ZmWUS1* activity through a receptor-like protein FEA3 (Je et al., 2016), as well as via FEA2 and *ZmCORYNE* (*ZmCRN*) (Je et al., 2018). Notably, *ZmFCP1* is produced in and transported from emerging leaf primordia (Je et al., 2016).

Another program that plays an indispensable role in maize ear development is the FEA4-MALE STERILE CONVERTED ANTHER1 (MSCA1) module (Fig. 2B). *FEA4* encodes a bZIP transcription factor and is expressed throughout the female IM (Pautler et al., 2015; Yang et al., 2021). *fea4* mutant ears are massively fasciated and disorganized (Pautler et al., 2015). *FEA4* interacts with and is a major target of MSCA1, a glutaredoxin (Yang et al., 2015; Yang et al., 2021). MSCA1 acts redundantly with two of its paralogs, *ZmGRX2* and *ZmGRX5* (Yang et al., 2021). In the *m sca1 zmgrx2 zmgrx5* triple mutant, *FEA4* is in a potentially oxidized dimer state and has higher access to its target promoters. As a result, the ear size is significantly reduced (Yang et al., 2021). Interestingly, both *FEA4* and *MSCA1* are bound by *ZmWUS1* and *ZmWUS2*, suggesting that the *FEA4*-*MSCA1* module and the WUS-CLV loop might modulate ear development cooperatively (Dong et al., 2023).

SPM initiation and development require SBP-box family transcription factors encoded by *tassel sheath4* (*tsh4*) and its

Table 1 Key genes regulating maize ear development

Stage	Gene	Gene product	Expression domain	Mutant phenotype	References
IM	<i>ZmWUSCHEL1</i> (<i>ZmWUS1</i>)	WOX TF	organizing center of the IM	ball-shaped ears with barren patches caused by ectopic expression of an additional copy	Nardmann & Werr (2006); Je et al. (2016); Chen et al. (2021), (2024)
	<i>ZmWUSCHEL2</i> (<i>ZmWUS2</i>)	WOX TF	throughout the IM	not observed	Nardmann & Werr (2006)
	<i>ZmCLAVATA3/EMBRYO SURROUNDING REGION-RELATED7</i> (<i>ZmCLE7</i>)	CLE peptide	central zone of the IM	enlarged, fasciated, ridged, bifurcated, and otherwise misshapen ears often with extra and/or disorganized kernel rows	Je et al. (2016); Rodriguez-Leal et al. (2019); Chen et al. (2021); Liu et al. (2021)
	<i>thick tassel dwarf1</i> (<i>tdt1</i>)	LRR receptor-like kinase	three outermost layers of the IM; emerging SPMs	same as above	Bommert et al. (2005); Lunde & Hake (2009)
	<i>fasciated ear2</i> (<i>fea2</i>)	LRR receptor-like protein	immature ears	same as above	Taguchi-Shiobara et al. (2001); Bommert et al. (2013a); Je et al. (2018)
	<i>COMPACT PLANT2</i> (<i>CT2</i>)	α of a heterotrimeric GTP-binding protein	outer layers of the IM	same as above	Bommert et al. (2013b)
	<i>ZmFON2-LIKE CLE PROTEIN1</i> (<i>ZmFCP1</i>)	CLE peptide	emerging leaf primordia	fasciated ears with increased kernel rows caused by gene mutation or silencing; inhibited IM growth	Je et al. (2016); Liu et al. (2021)
	<i>fasciated ear3</i> (<i>fea3</i>)	LRR receptor-like protein	lower layers of the IM	fasciated ears with additional and disorganized kernel rows	Je et al. (2016)
	<i>ZmCORYNE</i> (<i>ZmCRN</i>)	transmembrane pseudokinase	throughout the SAM	same as above	Je et al. (2018)
	<i>FASCIATED EAR4</i> (<i>FEA4</i>)	bZIP TF	throughout the IM	same as above	Pautler et al. (2015); Yang et al. (2021)
SPM	<i>MALE STERILE CONVERTED ANTH1</i> (<i>MSCA1</i>)	glutaredoxin	central region of the IM	triple mutants have extremely small ears	Yang et al. (2015)
	<i>Zea mays Glutaredoxins2/5</i> (<i>ZmGRX2/5</i>)	glutaredoxin	central region of the IM	same as above	Yang et al. (2015)
	<i>tasselsheath4</i> (<i>tsh4</i>)	SBP-box TF	SPM anlage	fewer SPMs; de-repressed bracts; distorted kernel rows	Chuck et al. (2010); Dong et al. (2024)
	<i>unbranched2/3</i> (<i>ub2/3</i>)	SBP-box TF	overlaps completely with TSH4 protein	double mutants have extra SPM ranks; triple mutants combining <i>tsh4</i> have early terminated IM	Chuck et al. (2014)
	<i>Corngrus1</i> (<i>Cgt1</i>)	miR156	leaf primordia	de-repressed bracts caused by ectopic expression of an additional copy	Chuck et al. (2007a)

Continued

Table 1 Continued

Stage	Gene	Gene product	Expression domain	Mutant phenotype	References
	<i>ramosa1 (ra1)</i>	C2H2 zinc finger TF	surrounding the base of the SPM	additional branching and/or loss of determinacy in the SPM	Vollbrecht et al. (2005)
	<i>RAMOSA1 ENHANCER LOCUS2 (REL2)</i>	TOPLESS transcriptional co-repressor	same as above	same as above	Gallavotti et al. (2010)
	<i>ramosa2 (ra2)</i>	LOB TF	same as above	same as above	Vollbrecht et al. (2005); Bortiri et al. (2006)
	<i>ramosa3 (ra3)</i>	trehalose-6-phosphate phosphatase	same as above	same as above	Sato-Nagasawa et al. (2006); Claeys et al. (2019)
SM	<i>indeterminate spikelet1 (ids1)</i>	AP2 TF	throughout the SPM and SM	extra florets	Chuck et al. (1998)
	<i>sister of indeterminate spikelet1 (sid1)</i>	AP2 TF	throughout the SPM and SM	double mutants combining <i>ids1</i> have a bare rachis and reduced SPM ranks	Chuck et al. (2008)
	<i>branched silkless1 (bd1)</i>	AP2 TF	a semicircular domain at the glume and SM junction	SMs replaced by branches producing SPMs indeterminately	Colombo et al. (1998); Chuck et al. (2002)
	<i>tasselseed4 (ts4)</i>	miR172e	base of the SPM and SM	extra florets	Chuck et al. (2007b)

AP2, APETALA2; bZIP, basic leucine zipper; GTP, guanosine triphosphate; IM, inflorescence meristem; LOB, LATERAL ORGAN BOUNDARIES; LRR, leucine-rich repeat; SAM, shoot apical meristem; SBP, SQUAMOSA-promoter binding protein; SM, spikelet meristem; SPM, spikelet pair meristem; TF, transcription factor; WOX, WUSCHEL-related homeobox.

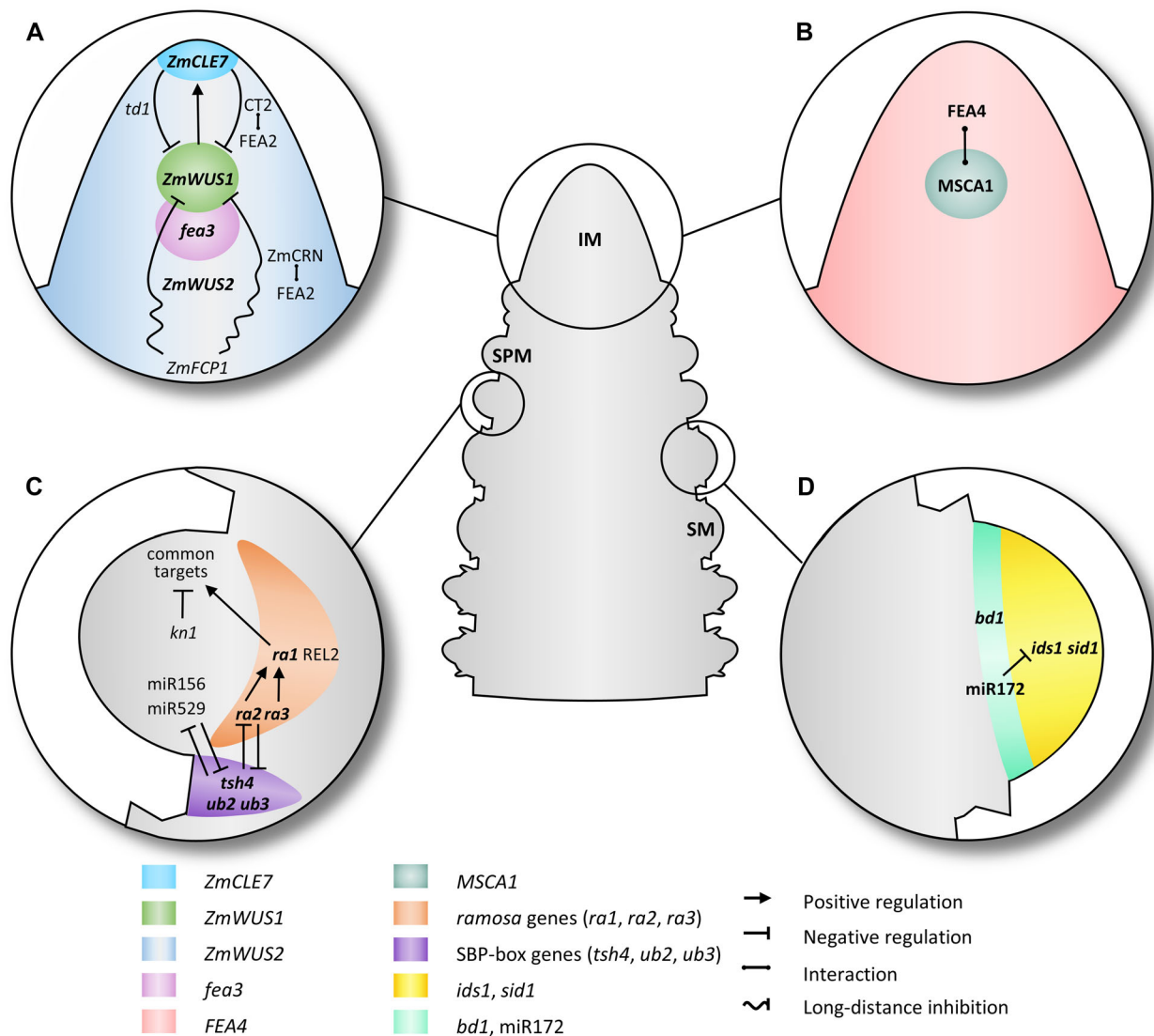


Fig. 2. Gene regulatory networks in developing maize ear. **A**, WUS-CLV feedback loop controls the maintenance of inflorescence meristem. Note that *ZmFCP1* is transported from leaf primordia. **B**, FEA4-MSCA1 module regulates the function of IM. **C**, *ra* pathway mediates the determinacy of spikelet pair meristem. **D**, *ids1* and *sid1* function together to govern the determinacy of spikelet meristem. Bold letters and colored shadows represent known gene expression domains. FM, floral meristem; IM, inflorescence meristem; SM, spikelet meristem; SPM, spikelet pair meristem.

paralogs *unbranched2* (*ub2*) and *ub3* (Chuck et al., 2010, 2014; Dong et al., 2024) (Fig. 2C). *tsh4* plays a major role in establishing the SPM anlage, and the *tsh4* mutant has reduced SPM formation (Chuck et al., 2010). Despite its crucial function during SPM initiation, *tsh4* is excluded from the developing SPM by several microRNAs and SPM determinacy genes through a double negative feedback mechanism (Chuck et al., 2007a, 2010; Dong et al., 2024). *tsh4*, *ub2*, and *ub3* also redundantly control the rate of SPM initiation (Chuck et al., 2014). In double and triple mutants, cells in the female IM are allocated to produce the SPM at a higher rate, causing the premature termination of the female IM (Chuck et al., 2014). Moreover, *ub2* and *ub3* modulate SPM number, as extra ranks are formed in double mutant ears (Chuck et al., 2014). This could be explained by the direct *ub3*

regulation of *ZmWUS1* and *ZmFCP1* expression levels (Du et al., 2017). *ub3* itself is, in turn, fine-tuned by a downstream intergenic major quantitative trait locus (QTL) for kernel row number (KRN) known as *KRN4* (Liu et al., 2015). *KRN4* functions as a distant enhancer of *ub3* via chromatin interactions with its promoter, and this is mediated by the recruitment of UB2-centered transcription complexes (Du et al., 2020).

SPM determinacy is mainly controlled by the *ramosa* (*ra*) pathway (Fig. 2C). *ra1* encodes a C2H2 zinc finger transcription factor and is expressed in a region surrounding the base of the SPM, where it imposes SPM determinacy non-cell-autonomously (Vollbrecht et al., 2005). The absence or reduction of *ra1* delays SPM determinacy, resulting in the formation of branches or spikelet multimers instead of

spikelet pairs (Vollbrecht et al., 2005). *ra1* promotes branch determinacy either via the integration with the regulatory network of *knotted1* (*kn1*) (Eveland et al., 2014), a master regulator of maize meristem maintenance (Jackson et al., 1994; Kerstetter et al., 1997), or by forming a repressor complex with a co-repressor named RAMOSA1 ENHANCER LOCUS2 (*REL2*) (Gallavotti et al., 2010). *ra2* and *ra3* act upstream of *ra1* in two different manners. *ra2* is expressed earlier than and transiently overlaps with *ra1* in the SPM anlage, and may be involved in *ra1* transcription activation (Vollbrecht et al., 2005; Bortiri et al., 2006). *ra3* co-localizes with and regulates *ra1* by a mechanism involving trehalose-6-phosphate phosphatase (Satoh-Nagasawa et al., 2006; Claeys et al., 2019).

SM determinacy is partly regulated by *indeterminate spikelet1* (*ids1*) (Chuck et al., 1998) and its paralog sister of *indeterminate spikelet1* (*sid1*) (Chuck et al., 2008) (Fig. 2D). Both genes encode AP2 family transcription factors targeted by miR172e, which is produced by *tasselseed4* (*ts4*) (Chuck et al., 2007b). *ids1* and *sid1* are expressed within the SPM and SM (Chuck et al., 1998, 2008), whereas *ts4* is expressed complementarily at their bases (Chuck et al., 2007b). In *ids1* loss-of-function mutant ears, several extra florets are initiated from the SM in a distichous pattern (Chuck et al., 1998, 2007b). In contrast, in *ts4* mutant ears, in which *ids1* is mis-expressed, extra florets are formed randomly (Irish, 1997; Chuck et al., 2007b). In *ids1 sid1* double mutant ears, the SM fails to transition to the FM fate, resulting in a bare rachis (Chuck et al., 2008). SM determinacy is also regulated by another AP2 family transcription factor encoded by *branched silkless1* (*bdl*) (Colombo et al., 1998; Chuck et al., 2002) (Fig. 2D). *bdl* is not expressed in the SM itself but instead in a semicircular domain at the glume and SM junction (Chuck et al., 2002). In *bdl* mutant ears, the SM is often replaced by an indeterminate axis producing SPMs in a distichous phyllotaxy (Colombo et al., 1998; Chuck et al., 2002). After the SM giving rise to the lower and upper FMs, their differential fate is regulated by miR172e, as well as hormones, sugar availability, energy homeostasis, and cell wall composition (Yang et al., 2022; Zheng et al., 2025).

3 Morphological and developmental differences between teosinte and maize ears

Despite a century-long debate (Harshberger, 1896; Collins, 1919, 1921; Beadle, 1932a, 1932b, 1939, 1972; Emerson & Beadle, 1932; Mangelsdorf & Reeves, 1938; Mangelsdorf, 1958, 1974; Doebley, 2001), compelling evidence supports a model that, maize was initially domesticated from an annual lowland teosinte *Z. mays* ssp. *parviglumis* in the Balsas River basin of southwest Mexico, followed by an admixture with another annual highland teosinte *Z. mays* ssp. *mexicana* (Doebley et al., 1984; Piperno & Flannery, 2001; Matsuoka et al., 2002; van Heerwaarden et al., 2011; Yang et al., 2023). Unlike maize, teosinte exhibits elongated lateral branches and often has tillers. Its tassels (male inflorescences) are formed at the terminal of the main stem and every long lateral branch and tiller, while ears are also in leaf axils. Teosinte has multiple

ear-bearing branches and tillers, and a teosinte leaf axil often has an ear cluster resulting from highly condensed higher-order branches, rendering teosinte much more prolific than maize.

At the individual ear level, the most readily apparent difference between teosinte and maize is the size. A mature teosinte ear is significantly smaller in both length and girth. Each teosinte ear has only two opposite kernel ranks, with each producing a single row (Fig. 3A); whereas a maize ear has at least four ranks, with each forming two rows (Fig. 1A). As a result, the number of kernels per ear is drastically increased from teosinte (5–12) to maize (>500). This morphological difference results from their divergent IM developmental processes. A teosinte IM is much smaller than a maize IM, thus causing a smaller number of SPMs to be initiated. Moreover, teosinte pedicellate SMs abort during the SM to FM transition, leaving only sessile SMs (Fig. 3B) (Sundberg & Orr, 1990). Another major distinction lies in fruitcasing. Each teosinte kernel is encased by a hardened outer glume on the outside, and a segment of the invaginated rachis (cupule) to the inside (Doebley, 1992; Doebley et al., 1995; White & Doebley, 1998). Such a fruitcasing structure that protects the kernel is not found in maize (Doebley et al., 1990). Upon maturity, a teosinte ear develops abscission layers and, as a result, disarticulates and shatters, giving rise to individual fruitcases. The integrity of maize ear is maintained as individual maize kernels are attached to the enlarged central fibrous rachis till harvest.

4 Mechanisms governing the origin and domestication of maize ear

Although recent studies have shed light on the process of maize domestication, the origin of maize ear is still not fully understood. Due to the stunning differences between teosinte and maize ears, a plethora of hypotheses or theories regarding this myth have been proposed historically (Iltis, 2000). The Orthodox Teosinte Hypothesis (OTH) proposes that maize ear was directly derived from teosinte ear via a series of morphological transformation events, including the enlargement and elongation of the ear, the reactivation of pedicellate spikelets, and the loss of disarticulation (Harshberger, 1907; Beadle, 1939; Iltis, 1971, 2000). Nevertheless, the OTH cannot explain differences in terms of ear position (primary branch vs. secondary and higher-order branches) and number (solitary vs. clustered) between maize and teosinte. To reconcile these major discrepancies, the Catastrophic Sexual Transmutation Theory (CSTT) posits that maize ear is transformed from a feminized and condensed central spike of a teosinte tassel that tips a primary branch (Doebley & Iltis, 1980; Iltis, 1983, 1987; Iltis & Doebley, 1984). Although mixed bisexual inflorescences tipping lateral branches in teosinte can often be observed, this may be merely resulting from smut infection, insufficient light, or cold (Iltis, 2000).

Neither the OTH nor the CSTT could satisfactorily account for the origin of maize ear. Inspired by the homeosis concept (Sattler, 1988), these hypotheses have been further synthesized as the Sexual Translocation Theory (STLT), which postulates that maize ear is the result of a series of

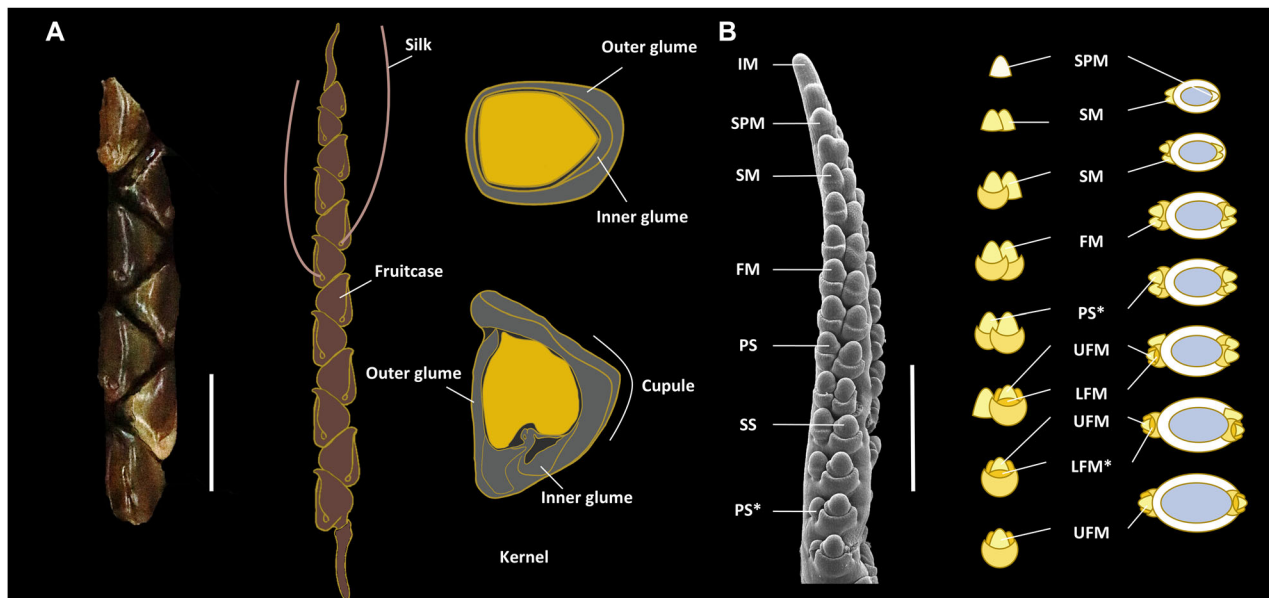


Fig. 3. Development of teosinte ear. **A**, Real and cartoon images of a mature teosinte ear, with cross and longitudinal sections of a kernel. The false grey color of glume is for better visualization. Bar = 1 cm. **B**, Scanning electron micrograph of a developing teosinte ear, with schematic meristem identities. Asterisks indicate aborting organs. White and grey indicate different organ attachment levels. Bar = 500 μ m. FM, floral meristem; IM, inflorescence meristem; LFM, lower floral meristem; PS, pedicellate spikelet; SM, spikelet meristem; SPM, spikelet pair meristem; SS, sessile spikelet; UFM, upper floral meristem.

incremental and somewhat stepwise hypothetical key transitions during domestication (Illitis, 2000). These key transitions are: 1) reduction of glumes and liberation of kernels; 2) reactivation of pedicellate spikelets; 3) dominance of only one ear and suppression of all other ears within each ear cluster, such that an ear cluster becomes an ear; 4) dominance of only one or two ears per plant; and 5) “translocation” from male to female on the primary and higher-order branches as a result of branch shortening and subsequent crossing of the hypothetical “sexual threshold zone” (Illitis, 2000).

The STLT offers a tempting framework to address the enigma of the origin of maize ear. Nevertheless, how the genetic programs could have been altered during these key transitions is not integrated. Based on the phenotypic frequency of F_2 offspring from a maize-teosinte cross, it has been suggested that as few as five major loci could have converted teosinte into maize (Beadle, 1972). Over the past decades, several QTLs and key candidate domestication genes affecting maize architecture and ear morphology have been discovered (Doebley et al., 1990; Doebley & Stec, 1991; Doebley, 1992; Bombliés & Doebley, 2006). One key domestication gene is *teosinte branched1* (*tb1*), a TCP family transcription factor that underlies a large effect QTL responsible for distinguishing short branches tipped by ears from long branches tipped by tassels (Doebley & Stec, 1991, 1993; Doebley et al., 1995, 1997). Mutation of *tb1* in maize leads to the emergence of long branches tipped by tassels and the outgrowth of tillers, phenotypes resembling teosinte (Burnham, 1959), implying that *tb1* inhibits the outgrowth of lateral branches and facilitates the formation of ears over tassels (Doebley et al., 1997). In ear primordia, the maize allele of *tb1* is expressed at approximately twice the level of the

teosinte allele, consistent with the hypothesis that earlier and higher expression of *tb1* is selected for during domestication (Doebley et al., 1997). This expression difference is at least due to a single nucleotide polymorphism (SNP) in the 5' non-transcribed region (Wang et al., 1999), as well as a *Hopscotch* transposon insertion in a regulatory region ~58 kb upstream of the *tb1* promoter (Clark et al., 2006; Studer et al., 2011). Nucleotide diversity analysis also indicates that sequences reaching as far as 90 kb upstream of *tb1* are targets of a selective sweep during maize domestication (Clark et al., 2004). Notably, *tb1* is expressed at the base of the pedicellate spikelet in developing teosinte ears. Such an expression pattern is not observed in maize ears, suggesting that suppressed *tb1* could account for the reactivation of pedicellate spikelet in maize ears (Hubbard et al., 2002).

Using another mapping population, *prol1.1* has been detected as a large effect QTL for prolificacy (number of ears per plant) (Wills et al., 2013). *prol1.1* maps to the promoter of *grassy tillers1* (*gt1*), which encodes an HD-ZIP family transcription factor involved in the regulation of plant architecture and ear clustering (Whipple et al., 2011; Wills et al., 2013). In maize, *gt1* is expressed in axillary buds (Whipple et al., 2011). Plants in the mapping population that harbor homozygous teosinte *prol1.1* typically generate ear clusters at leaf axils and thus more ears per plant, and lines bearing the maize allele have upregulated *gt1* expression in primary branches (Wills et al., 2013). These findings imply that the allelic differences at *prol1.1* involve modifications in a *cis*-regulatory element that lead to increased *gt1* expression in the primary branch of maize, which in turn inhibits the initiation of secondary and higher-order ear buds (Wills et al., 2013). Population genetic analyses further reveal

positive selection on the maize allele of *prol1.1*, causing a partial sweep that fixed the maize allele during domestication (Wills et al., 2013).

The active copy number of *Zea floricaula/leafy2* (*zfl2*) and its paralog *zfl1* is significantly associated with numerous ear traits, with *zfl2* being quantitatively correlated with the formation of multiple ranks (Bomblies et al., 2003; Bomblies & Doebley, 2006). Both genes map close to previously identified QTLs for ear rank number (Doebley & Stec, 1991, 1993; Bomblies et al., 2003). A complementation test has revealed that teosinte *zfl2* alleles in hybrid populations might be equivalent to loss-of-function maize *zfl2* alleles in terms of their effect on rank number (Bomblies & Doebley, 2006). *ids1* also affects ear rank number besides its function in governing SM determinacy (Wang et al., 2019). Elevated *ids1* expression increases ear rank number, and this may involve altered expression of several IM regulators such as *fea3* and *FEA4* (Wang et al., 2019). More recently, *KRN2* has also been recognized as playing an important role in regulating KRN. Selection upon its upstream noncoding regions results in reduced expression and increased KRN during domestication (Chen et al., 2022). *KRN2* encodes a WD40 protein and negatively regulates KRN by interacting with a protein of unknown function (Chen et al., 2022). Maize *KRN2* knockout lines produce more kernel rows as compared to the wild type without any noticeable trade-offs in other agronomic traits (Chen et al., 2022).

teosinte *glume architecture1* (*tga1*) underlies the major differences between kernel-encasing hardened glumes of teosinte, and soft and reduced glumes that expose kernels of maize (Doebley & Stec, 1991, 1993; Dorweiler et al., 1993). The maize allele of *tga1* in the teosinte background results in a less invaginated and shorter rachis and an upward oriented glume, which becomes duller due to disrupted silica deposition; in contrast, the teosinte allele of *tga1* in the maize background causes elongation of the rachis, rounding of the glume, and intensified silica deposition and additional layers of lignified cells in the glume (Dorweiler et al., 1993; Dorweiler & Doebley, 1997). *tga1* encodes an SBP-box family transcription factor targeted by miR156 (Wang et al., 2005; Chuck et al., 2007a). Despite similar expression patterns and levels in SPM, SM, and glume between teosinte and maize, a fixed SNP in the maize *tga1* coding region causes an amino acid substitution (K6N) (Wang et al., 2005). This results in the formation of more stable TGA1 protein dimers and renders maize TGA1 acting more as a repressor (Wang et al., 2015). Genetic and molecular evidence further support a scenario that TGA1 directly interacts with components of the strigolactone signaling pathway, which mediates TGA1 protein availability by sequestration (Guan et al., 2023).

Given the complexity of the domestication of maize ear, selection is likely to operate upon a genetic regulatory network, rather than discrete, independently acting loci. Indeed, it has been demonstrated that *tb1* directly targets a suite of domestication genes described above, including *gt1*, *tga1*, as well as itself (Whipple et al., 2011; Studer et al., 2017; Dong et al., 2019). Consistent with its role as a repressor of organ growth, *tb1* also mediates axillary branching suppression through the direct activation of *tassel replace upper ears1* (*tru1*), which represses internode elongation and controls sex identity in lateral branches (Dong et al., 2017), and via the

inhibition of two core cell cycle genes required for cell division (Studer et al., 2017). Moreover, TSH4, UB2, and UB3 bind and activate *tb1*, *tga1*, and *tru1* (Liu et al., 2021; Dong et al., 2024), further complicating this genetic regulatory network.

Combining recent molecular evidence with the framework of the classic STLT, we propose a model for the origin of maize ear (Fig. 4). During domestication, *tb1* expression was enhanced, possibly partially affected by *tsh4*, *ub2*, and *ub3*. Elevated *tb1* expression, in turn, activated *gt1* and *tru1*. *tb1*, *gt1*, and *tru1* collectively repressed the initiation, outgrowth, and internode elongation of lateral branches. As a result, only one or two shortened and condensed primary branch(es) tipped by only one terminal inflorescence remained. Subsequently, as the remaining primary branch was continuously shortened and its tipping inflorescence crossed the hypothetical “sexual threshold zone”, the inflorescence gradually transitioned from a tassel to an ear under the regulation of *tru1*. *tb1* also targets *tga1* and the gain-of-function *tga1* transformed hard glumes into soft and reduced alternatives, exposing seeds from their protective fruitcases. Moreover, downregulation of *tb1* in the pedicellate spikelet led to its reactivation, effectively doubling the spikelet number. Multiple genes including *zfl2*, *ids1*, and *KRN2* modified the activity of the WUS-CLV regulation loop, either directly or indirectly, resulting in a balanced increase of ear size and further KRN rise.

5 Perspective

Over the past decades, building upon classic genetics, the ever-advancing molecular technology and functional genomics have significantly deepened our understanding of the mechanisms underlying the development and domestication of maize ear. Studies have also uncovered a complex gene regulatory network and how this network has been rewired during maize domestication (Ricci et al., 2019; Marand et al., 2021, 2025). However, there remain several outstanding questions. For example, the molecular mechanisms governing teosinte ear formation remain unexplored, thus hindering a direct and effective comparison between teosinte and maize. In addition, a detailed spatiotemporal transcriptome atlas is not yet available. To tackle these questions, integrative novel techniques and interdisciplinary strategies, such as multi-omics analyses and computational modeling, could provide key insights.

Leveraging single-cell RNA sequencing (scRNA-seq), specific cell type markers are identified and functionally validated in developing maize ears (Xu et al., 2021). Integrating scRNA-seq with chromatin immunoprecipitation sequencing (ChIP-seq) and assay for transposase accessible chromatin with sequencing (ATAC-seq) allows the exploration of genetic redundancy and transcriptional networks beyond traditional forward genetics (Xu et al., 2021). More recently, spatial enhanced resolution omics-sequencing (Stereo-seq) further refines our understanding by mapping gene expression at the cellular resolution, revealing novel regulators in specific meristem niches (Wang et al., 2024). Future application of these techniques to teosinte, coupled with gene editing (Zobrist et al., 2024) and comparative

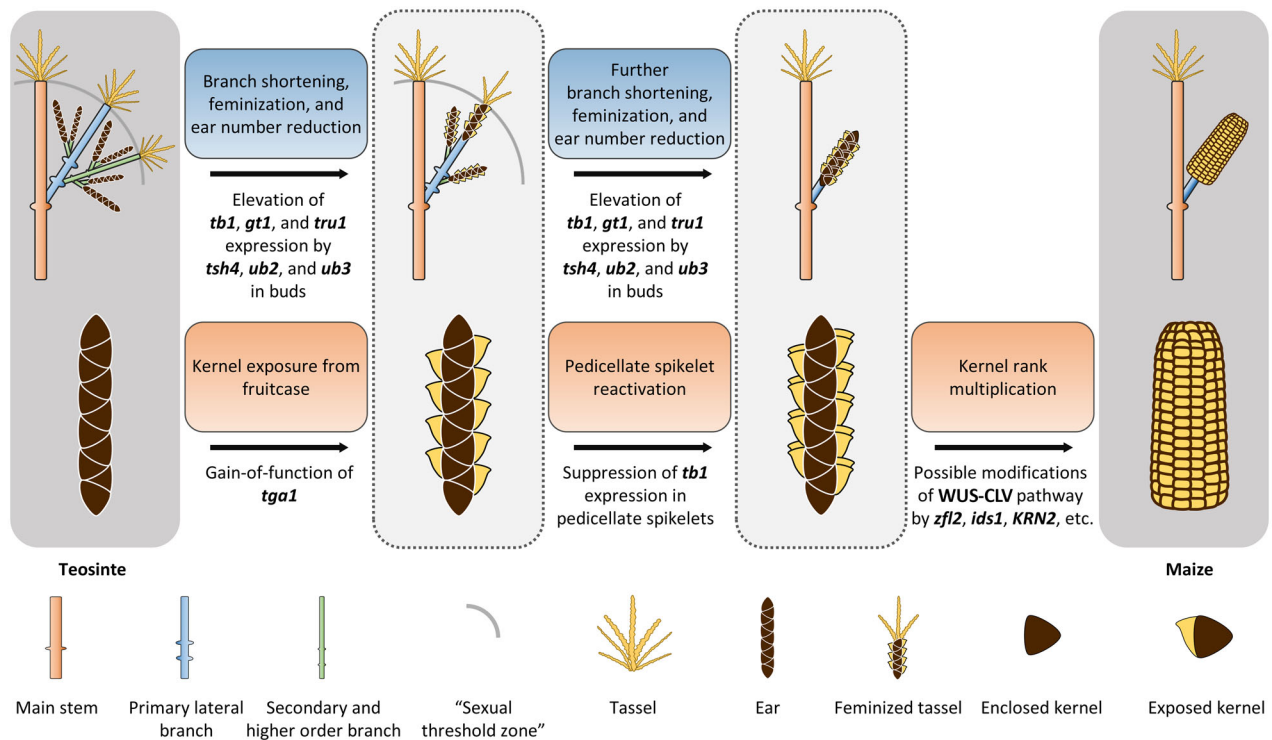


Fig. 4. A proposed model for the origin of maize ear. During domestication, elevated *tb1* enhanced *gt1* and *tru1* expression. Together, these genes suppressed the initiation, outgrowth, and internode elongation of lateral branches, leading to the reduction of the number of ears and the transition from a terminal tassel to a terminal ear on the primary branch. The gain-of-function *tga1* allele drove glume softening, facilitating seed exposure from protective fruitcases. Next, the activity of *tb1*, *gt1*, and *tru1* ultimately transformed the primary branch into a highly reduced branch with only an ear attached to its tip. In addition, the *tb1* downregulation reactivated pedicellate spikelets, thus doubling the number of spikelets. Furthermore, various genes such as *zfl2*, *ids1*, and *KRN2* might modify the activity of the WUS-CLV pathway, resulting in a balanced increase of ear size and further kernel row number rise. Dotted boxes indicate hypothetical intermediates.

analyses, could uncover spatiotemporal differences in developmental process, gene expression, and regulatory network between teosinte and maize.

Computational modeling has emerged as a powerful tool for decoding the developmental mechanisms of diverse inflorescence architectures, including the spike-like type (Prusinkiewicz et al., 2007). By incorporating gene regulatory networks into dynamic branching models, researchers can explain phenotypic variations in the inflorescence of Brassicaceae species and their mutants (Prusinkiewicz et al., 2007; Azpeitia et al., 2021). In the future, the integration of artificial intelligence (AI)-driven simulation offers a transformative strategy for research focusing on inflorescence, enabling *in silico* prediction and validation of maize ear models. This combined computational-experimental paradigm could not only deepen our understanding of the molecular mechanisms governing the development and domestication of maize ear but also accelerate the development of intelligent molecular breeding strategies for crop improvement.

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Conflicts of Interest

The authors declare no conflict of interest.

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